

UNIT-III

Two marks questions and Answers

1. What do you understand by moment of Inertia? (May 2015)

(i) It is the property of the body due to which it resists angular acceleration.

(ii) Moment of Inertia of the body is the sum of the products of the area and the square of its distance from the axis of rotation.

2. Define the Centroidal axis. (May 2015)

The axis which are passing through the centroid of the figure is called the Centroidal axis.

3. State parallel axis theorem. (Jan. 2016)

It states that the moment of Inertia of a body about any axis in the plane of a body is equal to the sum of the moment of Inertia about a parallel centroidal axis in the plane and the product of the area and square of the distance between the two axes.

$$I_{xx} = I_G + ah^2$$

4. Define radius of gyration. (Jan. 2016)

Radius of gyration:

It is defined as the distance at which the whole area of a body may be supposed to be concentrated with respect to an axis of reference.

$$K_{xx} = \sqrt{I_{xx}/A} ; K_{yy} = \sqrt{I_{yy}/A}$$

5. Describe the second moment of area. (May 2016)

Second Moment of area:

(i) It is the sum of the product of the area and the square of the distance from the reference axis.

(ii) It is also called as the area moment of inertia.

6. Describe the polar moment of inertia. (May 2016)

The polar moment of inertia is equal to the sum of area of moment of inertia about any two mutual perpendicular axis in its plane and intersecting on the polar axis.

$$I_{zz} = I_{xx} + I_{yy} \text{ is theorem}$$

7. Find the radius of gyration of a rectangular area of MI about its base $9 \times 10^4 \text{ cm}^4$ and cross sectional area 300 cm^2

Solution:

$$I_{xx} = 9 \times 10^4 \text{ cm}^4$$

$$A = 300 \text{ cm}^2$$

$$\text{Radius of gyration, } k_{xx} = \sqrt{\frac{I_{xx}}{A}}$$

$$= \sqrt{\frac{9 \times 10^4}{300}}$$

$$= 17.32 \text{ cm}$$

$$= \underline{\underline{173.2 \text{ mm}}}$$

8. State Pappus - Guldinus theorem. (May 2017)

Theorem 1:

The area of any surface of revolution is equal to the length of the generating curve times the distance travelled by the centroid of the generating curve in generating the area.

Theorem 2:

The volume of any body of revolution is equal to the generating area times the distance by the centroid of the generating area in generating volume.

9. Differentiate centre of gravity and Centroid, (Dec. 2017)

Centre of gravity:

It is a point at which the entire mass of the body is assumed to be concentrated.

Centroid:

It is a point at which the entire area of the figure is assumed to be concentrated.

10. Define product of Inertia and give its properties. (Dec. 2018)

The product of Inertia of an area is summation of product of inertia of its elements about two perpendicular reference axis (x and y axis)

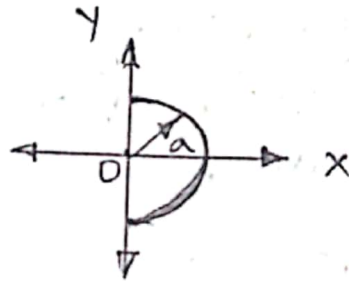
It is denoted by I_{xy} .

properties:

(i) The product of Inertia, I_{xy} , may be positive or negative or zero, depends on the location of the area to the axis.

(ii) If one or both x and y axis are symmetry, then the product of inertia is zero.

11. A semicircle of radius 'a', is defined in the first and fourth quadrant. Write down its Centroid
Solution: (Sep. 2020)



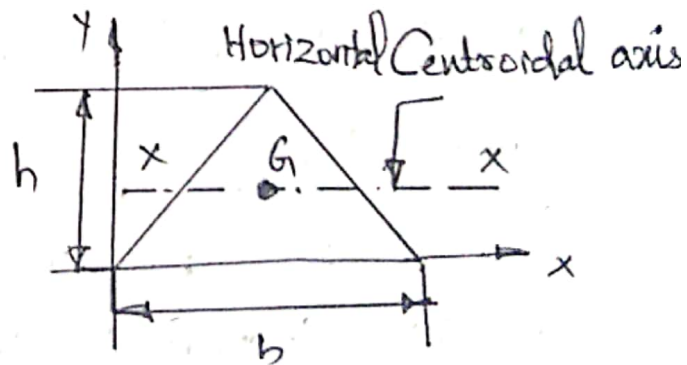
$$\bar{x} = \frac{2a}{\pi}$$

$$\bar{y} = 0$$

12. What is moment of Inertia of triangular lamina ($b \times h$) about its horizontal centroidal axis?

Solution:

$$\text{Moment of Inertia, } I_{xx} = \frac{bh^3}{36}$$

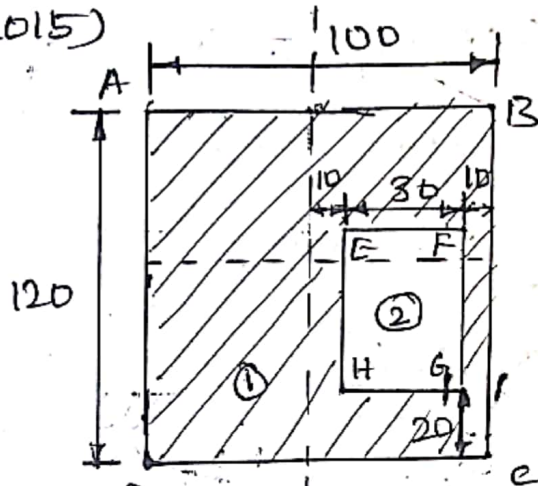


Unit III

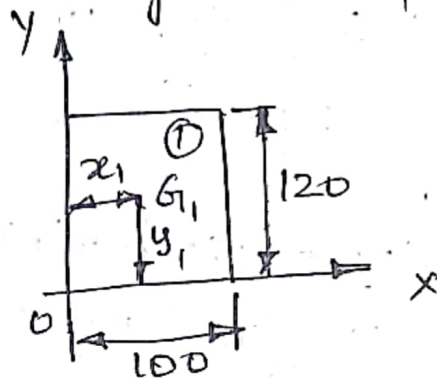
Centre of Gravity

1. From a rectangular lamina ABCD 100x120 mm a rectangular hole of 30x40 mm is cut as shown in fig. Find the C.G. of the remainder lamina. (May 2015)

Solution:



portion ①: Rectangle



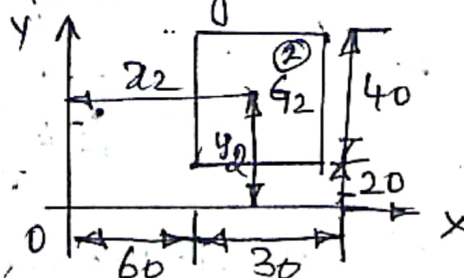
$$A_1 = l_1 \times b_1$$

$$= 120 \times 100 = 12000 \text{ mm}^2$$

$$x_1 = \frac{b_1}{2} = \frac{100}{2} = 50 \text{ mm}$$

$$y_1 = \frac{l_1}{2} = \frac{120}{2} = 60 \text{ mm}$$

portion ②: Rectangle (Removed portion)



$$A_2 = l_2 \times b_2$$

$$= 40 \times 30$$

$$= 1200 \text{ mm}^2$$

$$x_2 = 60 + \frac{b_2}{2}$$

$$= 60 + \frac{30}{2} = 75 \text{ mm}$$

$$y_2 = 20 + \frac{h_2}{2} = 20 + \frac{40}{2} = 40 \text{ mm}$$

$$\bar{x} = \frac{(a_1 x_1) - (a_2 x_2)}{a_1 - a_2}$$

$$= \frac{(12000 \times 50) - (1200 \times 75)}{12000 - 1200}$$

$$= \underline{\underline{47.22 \text{ mm}}}$$

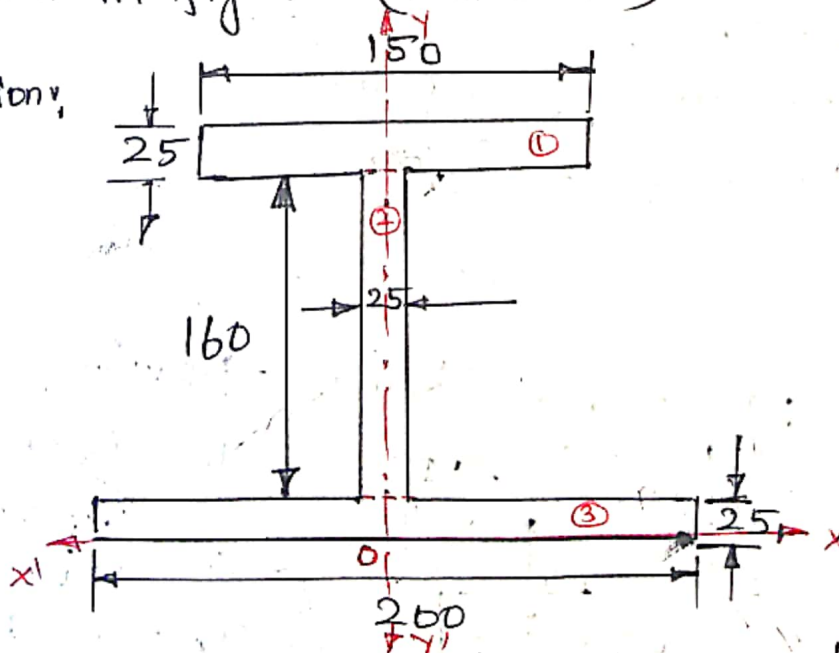
$$\bar{y} = \frac{(a_1 y_1) - (a_2 y_2)}{a_1 - a_2}$$

$$= \frac{(12000 \times 60) - (1200 \times 40)}{12000 - 1200}$$

$$= 62.22 \text{ mm}$$

- ∴ Centre of the gravity of the lamina = (47.22 mm, 62.22 mm)
- 2) Find the Centre of gravity for the given 'I' shown in figure. (Jan. 2016)

Solution:



All dimensions are in mm

Rectangle ①:

$$\text{Area, } a_1 = a_1 \times l_1 = 150 \times 25 = 3750 \text{ mm}^2$$

$$y_1 = 25 + 160 + \frac{25}{2} = 197.5 \text{ mm}$$

Rectangle ②:

$$a_2 = l_2 \times b_2 = 160 \times 25 = 4000 \text{ mm}^2$$

$$y_2 = \frac{l_2}{2} + 25 = \frac{160}{2} + 25 = 105 \text{ mm}$$

Rectangle ③:

$$a_3 = l_3 \times b_3 = 200 \times 25 = 5000 \text{ mm}^2$$

$$y_3 = \frac{b_3}{2} = \frac{25}{2} = 12.5 \text{ mm}$$

$\bar{x} = 0$ (Symmetrical about y-y axis)

$$\bar{y} = \frac{a_1 y_1 + a_2 y_2 + a_3 y_3}{a_1 + a_2 + a_3}$$

$$= \frac{(3750 \times 197.5) + (4000 \times 105) + (5000 \times 12.5)}{3750 + 4000 + 5000}$$

$$= \underline{\underline{95.93 \text{ mm}}}$$

∴ Centre of gravity of the given 'I' section = $(0, 95.93 \text{ mm})$

3. Determine the Centroid for the given cut section shown in fig.

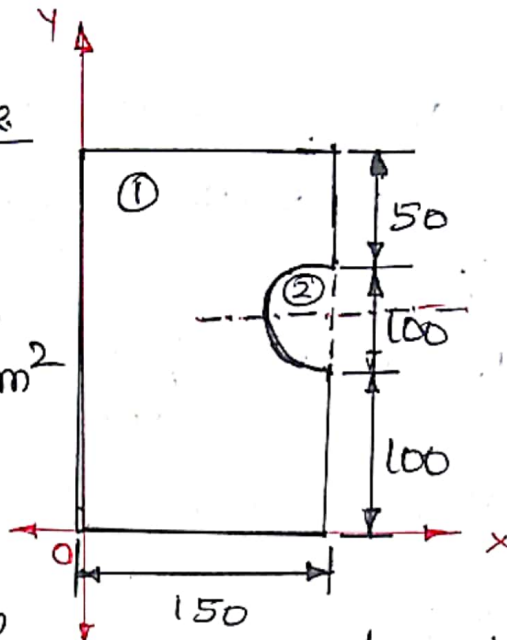
Solution:

portion ①: Rectangle

$$\begin{aligned}\text{Area, } a_1 &= l \times b, \\ &= 250 \times 150 \\ &= 37500 \text{ mm}^2\end{aligned}$$

$$\begin{aligned}x_1 &= \frac{b_1}{2} \\ &= \frac{150}{2} = 75 \text{ mm}\end{aligned}$$

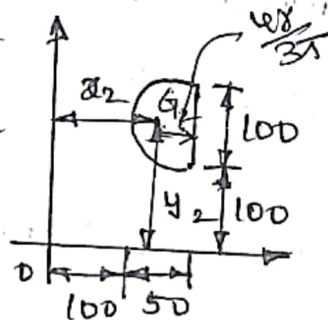
$$y_1 = \frac{l_1}{2} = \frac{250}{2} = 125 \text{ mm}$$



All dimensions are in mm

portion ②: Semi-circle (Removed portion)

$$\begin{aligned}\text{Area, } a_2 &= \frac{1}{2} \pi r^2 \\ &= \frac{1}{2} \pi (50)^2 \\ &= 3925 \text{ mm}^2\end{aligned}$$



$$\begin{aligned}x_2 &= 100 + \left(r - \frac{4r}{3\pi} \right) \\ &= 100 + \left(50 - \frac{4(50)}{3\pi} \right) \\ &= 128.76 \text{ mm}\end{aligned}$$

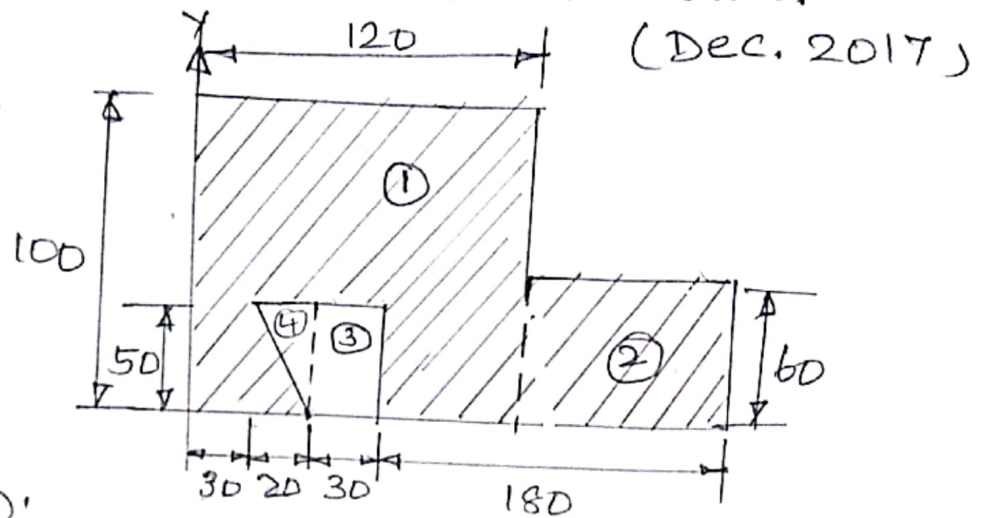
$$\begin{aligned}y_2 &= 100 + \frac{d}{2} \\ &= 100 + \frac{100}{2} \\ &= 150 \text{ mm}\end{aligned}$$

$$\begin{aligned}\bar{x} &= \frac{a_1 x_1 - a_2 x_2}{a_1 - a_2} = \frac{(37500 \times 75) - (3925 \times 128.76)}{37500 - 3925} \\ &= 68.71 \text{ mm}\end{aligned}$$

$$\begin{aligned}\bar{y} &= \frac{a_1 y_1 - a_2 y_2}{a_1 - a_2} = \frac{(37500 \times 125) - (3925 \times 150)}{37500 - 3925} \\ &= 122.07 \text{ mm}\end{aligned}$$

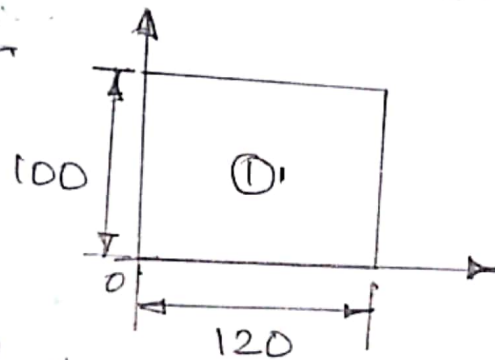
7. Determine the co-ordinates of the Centroid of the shaded area shown below.

Solutions:



Portion ①: Rectangle

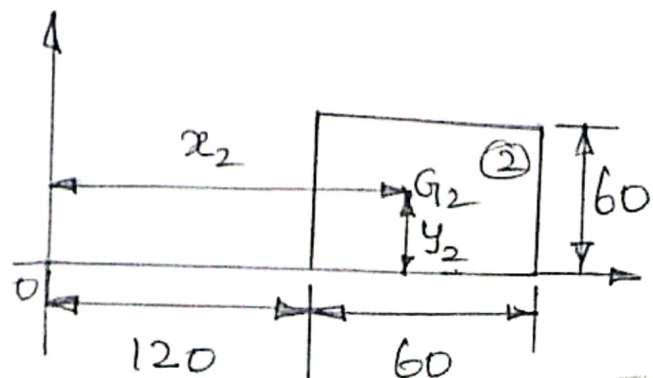
All dimensions are in mm



$$\begin{aligned} \text{Area, } a_1 &= l_1 \times b_1 \\ &= 120 \times 100 \\ &= 12000 \text{ mm}^2 \\ x_1 &= \frac{l_1}{2} = \frac{120}{2} = 60 \text{ mm} \\ y_1 &= \frac{b_1}{2} = \frac{100}{2} = 50 \text{ mm} \end{aligned}$$

Portion ②: Square

$$\begin{aligned} a_2 &= a_2^2 \\ &= 60 \times 60 \\ &= 3600 \text{ mm}^2 \\ x_2 &= \frac{a_2}{2} + 120 \\ &= \frac{60}{2} + 120 \\ &= 150 \text{ mm} \end{aligned}$$



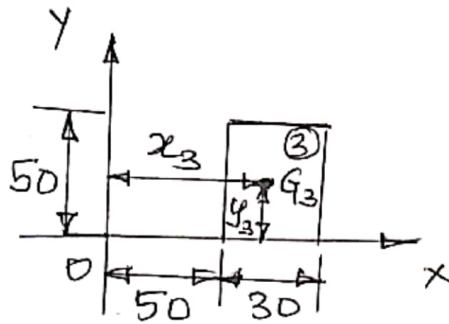
$$y_2 = \frac{a_2}{2} = \frac{60}{2} = 30 \text{ mm}$$

portion ③: Rectangle (Removed portion)

$$\begin{aligned} a_3 &= l_3 \times b_3 \\ &= 50 \times 30 \\ &= 1500 \text{ mm}^2 \end{aligned}$$

$$\begin{aligned} x_3 &= \frac{b_3}{2} + 50 \\ &= \frac{30}{2} + 50 = 65 \text{ mm} \end{aligned}$$

$$y_3 = \frac{l_3}{2} = \frac{50}{2} = 25 \text{ mm}$$

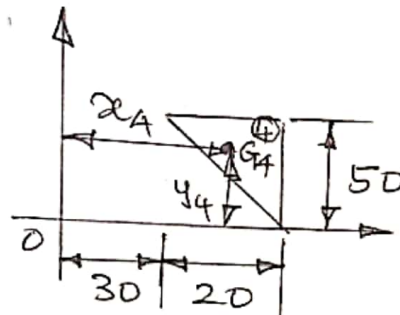


portion ④: Triangle (Removed portion)

$$\begin{aligned} a_4 &= \frac{1}{2} \times b_4 \times h_4 \\ &= \frac{1}{2} \times 20 \times 50 \\ &= 500 \text{ mm}^2 \end{aligned}$$

$$\begin{aligned} x_4 &= \left(b_4 - \frac{b_4}{3} \right) + 30 \\ &= \left(20 - \frac{20}{3} \right) + 30 \\ &= 43.33 \text{ mm} \end{aligned}$$

$$\begin{aligned} y_4 &= \left(h_4 - \frac{h_4}{3} \right) \\ &= 50 - \frac{50}{3} = 33.33 \text{ mm} \end{aligned}$$



$$\bar{x} = \frac{a_1 x_1 + a_2 x_2 - a_3 x_3 - a_4 x_4}{a_1 + a_2 - a_3 - a_4}$$

$$\begin{aligned} &= \frac{(12000 \times 60) + (3600 \times 150) - (1500 \times 65) - (500 \times 43.33)}{12000 + 3600 - 1500 - 500} \end{aligned}$$

$$= 83.88 \text{ mm}$$

$$\bar{y} = \frac{a_1 y_1 + a_2 y_2 - a_3 y_3 - a_4 y_4}{a_1 + a_2 - a_3 - a_4}$$

$$= \frac{(12000 \times 50) + (3600 \times 30) - (1500 \times 25) - (500 \times 33.33)}{12000 + 3600 - 1500 - 500}$$

$$= 48.07 \text{ mm}$$

5. Determine the co-ordinates of centroid of the shaded area shown in fig. (Dec. 2018)

Solution:

portion ①: Rectangle

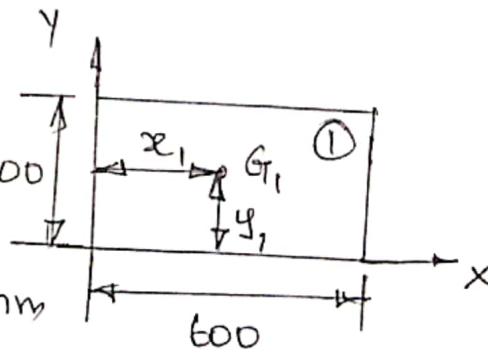
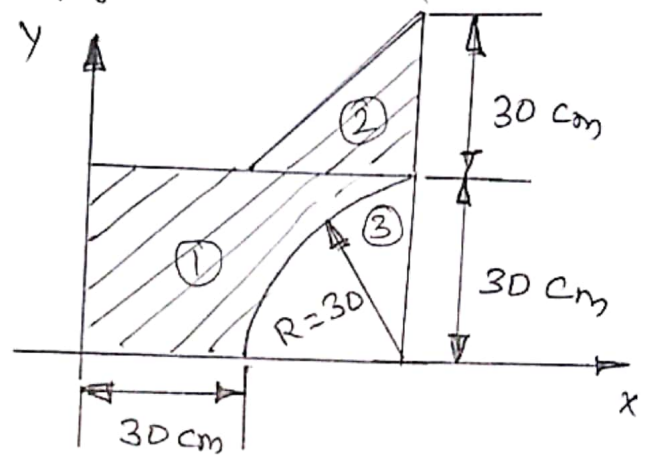
$$a_1 = l \times b,$$

$$= 600 \times 300$$

$$= 180000 \text{ mm}^2$$

$$x_1 = \frac{l_1}{2} = \frac{600}{2} = 300 \text{ mm}$$

$$y_1 = \frac{b_1}{2} = \frac{300}{2} = 150 \text{ mm}$$



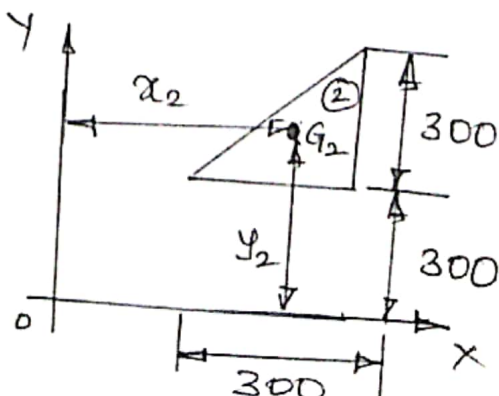
portion ②: Triangle

$$a_2 = \frac{1}{2} \times b_2 \times h_2$$

$$= \frac{1}{2} \times 300 \times 300$$

$$= 45000 \text{ mm}^2$$

$$x_2 = \frac{2}{3} b_2 + 300$$



$$= \frac{2}{3} \times 300 + 300$$

$$= 500 \text{ mm}$$

$$y_2 = \frac{h_2}{3} + 300$$

$$= \frac{300}{3} + 300$$

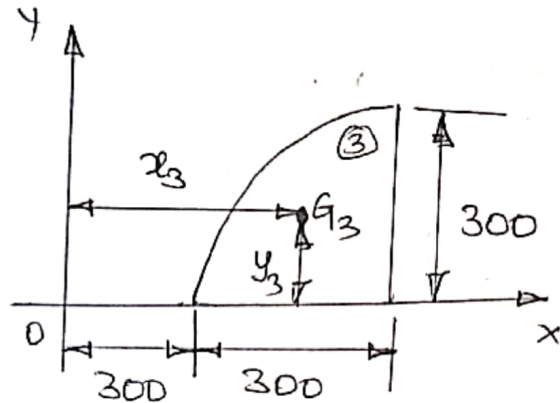
$$= 400 \text{ mm}$$

portion ③: Quadrant C (Removed portion)

$$a_3 = \frac{1}{A} \pi r^2$$

$$= \frac{1}{A} \times \pi (300)^2$$

$$= 70650 \text{ mm}^2$$



$$x_3 = \left(r - \frac{4r}{3\pi} \right) + 300$$

$$= \left(300 - \frac{4(300)}{3\pi} \right) + 300$$

$$= 472.62 \text{ mm}$$

$$y_3 = \frac{4r}{3\pi}$$

$$= \frac{4(300)}{3\pi}$$

$$= 127.38 \text{ mm}$$

$$\bar{x} = \frac{a_1 x_1 + a_2 x_2 - a_3 x_3}{a_1 + a_2 - a_3}$$

$$= \frac{(180000 \times 300) + (45000 \times 500) - (70650 \times 472.62)}{180000 + 45000 - 70650}$$

$$= 279.30 \text{ mm}$$

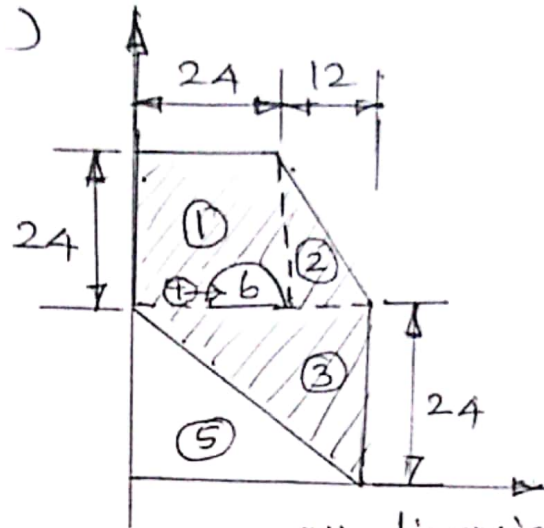
$$\bar{y} = \frac{a_1 y_1 + a_2 y_2 - a_3 y_3}{a_1 + a_2 - a_3}$$

$$= \frac{(180000 \times 150) + (45000 \times 400) - (70650 \times 127.38)}{180000 + 45000 - 70650}$$

$$= 233.34 \text{ mm}$$

6. Locate the centroid for the plane surface shown in fig. (Sep. 2020)

Solution:



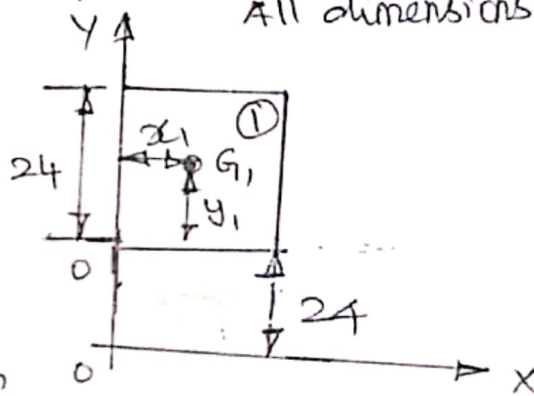
All dimensions are in mm

Portion ①: Square

$$\begin{aligned} \text{Area, } a_1 &= a_1^2 \\ &= (24)^2 \\ &= 576 \text{ mm}^2 \end{aligned}$$

$$x_1 = \frac{a_1}{2} = \frac{24}{2} = 12 \text{ mm}$$

$$y_1 = \frac{a_1}{2} + 24 = 12 + 24 = 36 \text{ mm}$$

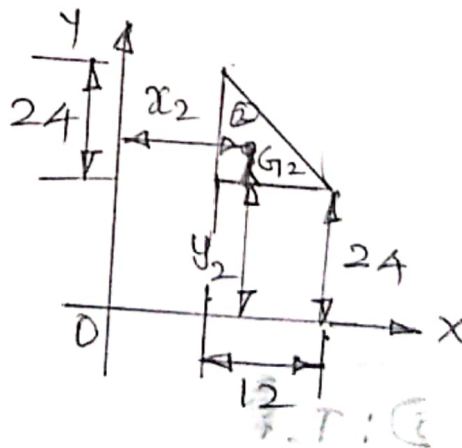


Portion ②: Triangle

$$\begin{aligned} a_2 &= \frac{1}{2} \times b_2 \times h_2 \\ &= \frac{1}{2} \times 12 \times 24 \\ &= 144 \text{ mm}^2 \end{aligned}$$

$$\begin{aligned} x_2 &= 24 + \frac{b_2}{3} \\ &= 24 + \frac{12}{3} = 28 \text{ mm} \end{aligned}$$

$$y_2 = 24 + \frac{h_2}{3} = 24 + \frac{24}{3} = 32 \text{ mm}$$



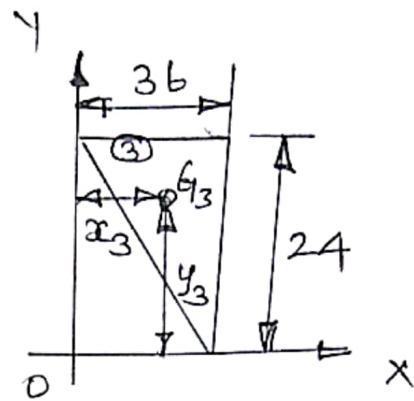
(4)

portion ③: Triangle

$$\begin{aligned}
 a_3 &= \frac{1}{2} \times b_3 \times h_3 \\
 &= \frac{1}{2} \times 36 \times 24 \\
 &= 432 \text{ mm}^2
 \end{aligned}$$

$$\begin{aligned}
 x_3 &= \left(b_3 - \frac{b_3}{3} \right) \\
 &= 36 - \frac{36}{3} = 24 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 y_3 &= \left(h_3 - \frac{h_3}{3} \right) \\
 &= 24 - \frac{24}{3} = 16 \text{ mm}
 \end{aligned}$$

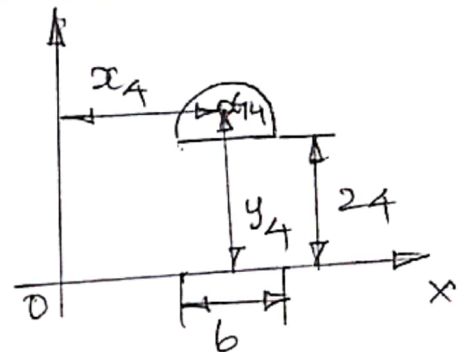


portion ④: Semicircle (Removed portion)

$$\begin{aligned}
 a_4 &= \frac{1}{2} \pi r^2 \\
 &= \frac{1}{2} \times \pi (3)^2 \\
 &= 14.13 \text{ mm}^2
 \end{aligned}$$

$$\begin{aligned}
 x_4 &= 18 + \frac{d_4}{2} = 18 + \frac{6}{2} \\
 &= 21 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 y_4 &= 24 + \frac{4r}{3\pi} \\
 &= 24 + \frac{4(3)}{3\pi} = 25.27 \text{ mm}
 \end{aligned}$$

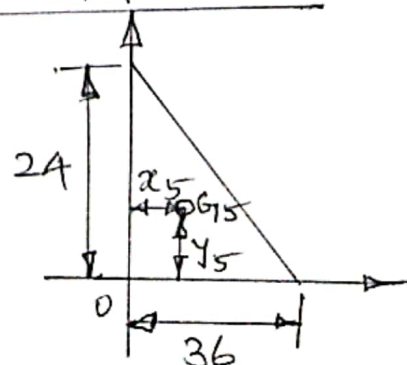


portion ⑤: Triangle (Removed portion)

$$\begin{aligned}
 a_5 &= \frac{1}{2} \times b_5 \times h_5 \\
 &= \frac{1}{2} \times 36 \times 24 \\
 &= 432 \text{ mm}^2
 \end{aligned}$$

$$x_5 = \frac{b_5}{3} = \frac{36}{3} = 12 \text{ mm}$$

$$y_5 = h_5 = 24 = 8 \text{ mm}$$



$$\bar{x} = \frac{a_1 x_1 + a_2 x_2 + a_3 x_3 - a_4 x_4 - a_5 x_5}{a_1 + a_2 + a_3 - a_4 - a_5}$$

$$= \frac{(576 \times 12) + (144 \times 28) + (432 \times 24) - (14.13 \times 21) - (432 \times 12)}{576 + 144 + 432 - 14.13 - 432}$$

$$= \frac{576 + 144 + 432 - 14.13 - 432}{24.42 \text{ mm}}$$

$$= 24.42 \text{ mm}$$

$$\bar{y} = \frac{a_1 y_1 + a_2 y_2 + a_3 y_3 - a_4 y_4 - a_5 y_5}{a_1 + a_2 + a_3 - a_4 - a_5}$$

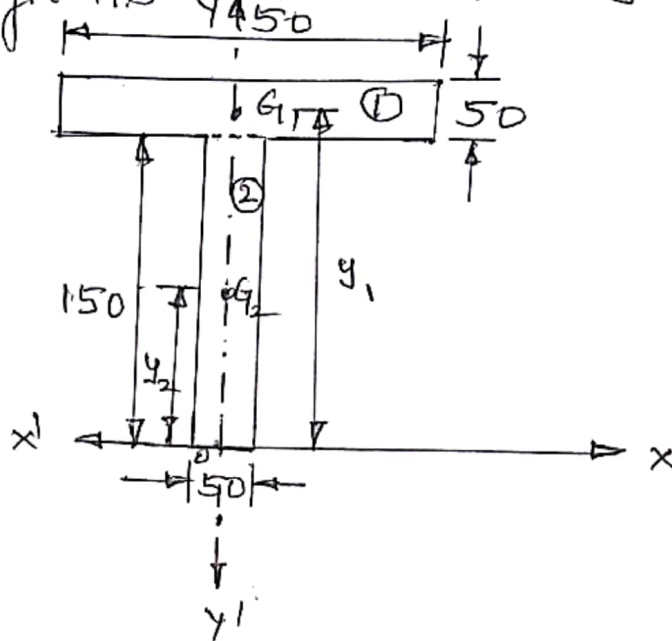
$$= \frac{(576 \times 36) + (144 \times 32) + (432 \times 16) - (14.13 \times 25.27) - (432 \times 8)}{576 + 144 + 432 - 14.13 - 432}$$

$$= \frac{576 + 144 + 432 - 14.13 - 432}{40.29 \text{ mm}}$$

$$= 40.29 \text{ mm}$$

Moment of Inertia

1. Find the area moment of inertia for the T section shown in fig. about x-x and y-y axis passing through its centroid. (May 2017)



Solution:

Rectangle ①:

$$a_1 = l_1 \times b_1 \\ = 150 \times 50 = 7500 \text{ mm}^2$$

$$y_1 = 150 + \frac{b_1}{2} = 150 + \frac{50}{2} \\ = 175 \text{ mm}$$

Rectangle ②:

$$a_2 = l_2 \times b_2 \\ = 150 \times 50 = 7500 \text{ mm}^2$$

$$y_2 = \frac{l_2}{2} = \frac{150}{2} = 75 \text{ mm}$$

$\bar{x} = 0$ ($\because$ symmetrical about $\bar{x}$ - $\bar{y}$ axis)

$$\bar{y} = \frac{a_1 y_1 + a_2 y_2}{a_1 + a_2} = \frac{(7500 \times 175) + (7500 \times 75)}{7500 + 7500} \\ = 125 \text{ mm}$$

Moment of inertia about x-x axis, (I_{xx})

$$I_{xx} = I_{xx_1} + I_{xx_2} \text{ ————— (1)}$$

$$\begin{aligned} I_{xx_1} &= IG_1 + a_1 h_1^2 \\ &= \frac{l_1 b_1^3}{12} + a_1 (y_1 - \bar{y})^2 \\ &= \frac{150(50)^3}{12} + 7500(175 - 125)^2 \\ &= 20.31 \times 10^6 \text{ mm}^4 \end{aligned}$$

$$\begin{aligned} I_{xx_2} &= IG_2 + a_2 h_2^2 \\ &= \frac{b_2 l_2^3}{12} + a_2 (y_2 - \bar{y})^2 \\ &= \frac{50(150)^3}{12} + 7500(175 - 75)^2 \\ &= 32.81 \times 10^6 \text{ mm}^4 \end{aligned}$$

sub. in (1),

$$\begin{aligned} \therefore I_{xx} &= 20.31 \times 10^6 + 32.81 \times 10^6 \\ &= 53.12 \times 10^6 \text{ mm}^4 \end{aligned}$$

Moment of inertia about y-y axis (I_{yy})

$$I_{yy} = I_{yy_1} + I_{yy_2} \text{ ————— (2)}$$

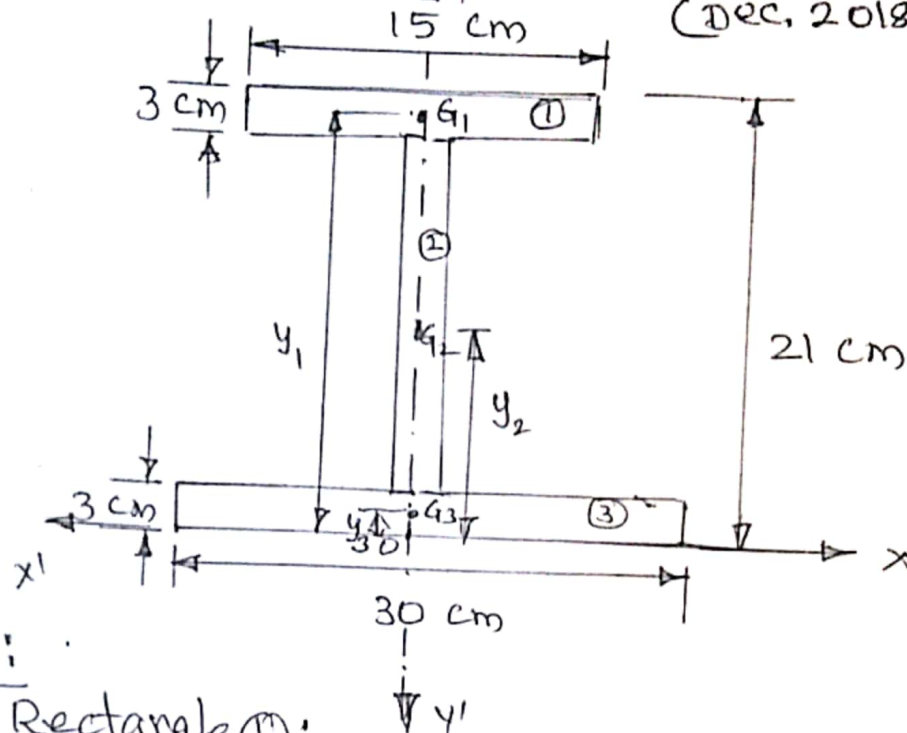
$$\begin{aligned} I_{yy_1} &= \frac{b_1 l_1^3}{12} + a_1 \bar{h}_1^2 \\ &= \frac{b_1 l_1^3}{12} + a_1 (\bar{x} - x_1)^2 \\ &= \frac{50(150)^3}{12} + a_1(0) = 14.06 \times 10^6 \text{ mm}^4 \end{aligned}$$

$$\begin{aligned} I_{yy_2} &= \frac{l_2 b_2^3}{12} + a_2 (\bar{x} - x_2)^2 \\ &= \frac{150(50)^3}{12} + a_2(0) = 1.56 \times 10^6 \text{ mm}^4 \end{aligned}$$

sub. in (2),

$$\begin{aligned} I_{yy} &= 14.06 \times 10^6 + 1.56 \times 10^6 \\ &= 15.62 \times 10^6 \text{ mm}^4 \end{aligned}$$

2. Find I_{xx} and I_{yy} through the centroid of 'I' section as shown in fig. All dimensions are in cm.
(Dec. 2018), (April 2018)



Solution:

Rectangle (1):

$$a_1 = l_1 \times b_1$$

$$= 150 \times 30 = 4500 \text{ mm}^2$$

$$y_1 = \frac{30}{2} + 180 = 195 \text{ mm}$$

Rectangle (2):

$$a_2 = 150 \times 30 = 4500 \text{ mm}^2$$

$$y_2 = \frac{150}{2} + 30 = 105 \text{ mm}$$

Rectangle (3):

$$a_3 = 300 \times 30 = 9000 \text{ mm}^2$$

$$y_3 = \frac{30}{2} = 15 \text{ mm}$$

$\bar{x} = 0$ ($\because$ symmetrical about $y-y$ axis)

$$\bar{y} = \frac{a_1 y_1 + a_2 y_2 + a_3 y_3}{a_1 + a_2 + a_3}$$

$$\bar{y} = \frac{(4500 \times 195) + (4500 \times 105) + (9000 \times 15)}{4500 + 4500 + 9000}$$

$$= \underline{\underline{82.5 \text{ mm}}}$$

Moment of Inertia about x-x axis (I_{xx}):

$$I_{xx} = I_{xx_1} + I_{xx_2} + I_{xx_3} \quad \text{--- (1)}$$

$$\begin{aligned} I_{xx_1} &= I_{G_1} + a_1 h_1^2 \\ &= \frac{l_1 b_1^3}{12} + a_1 (y_1 - \bar{y})^2 \\ &= \frac{150(30)^3}{12} + 4500(195 - 82.5)^2 \\ &= \underline{\underline{57.29 \times 10^6 \text{ mm}^4}} \end{aligned}$$

$$\begin{aligned} I_{xx_2} &= \frac{b_2 l_2^3}{12} + a_2 (y_2 - \bar{y})^2 \\ &= \frac{30(150)^3}{12} + 4500(105 - 82.5)^2 \\ &= \underline{\underline{10.715 \times 10^6 \text{ mm}^4}} \end{aligned}$$

$$\begin{aligned} I_{xx_3} &= \frac{l_3 b_3^3}{12} + a_3 (\bar{y} - y_3)^2 \\ &= \frac{300(30)^3}{12} + 9000(82.5 - 15)^2 \\ &= \underline{\underline{41.681 \times 10^6 \text{ mm}^4}} \end{aligned}$$

Sub. in (1),

$$\begin{aligned} I_{xx} &= 57.29 \times 10^6 + 10.715 \times 10^6 + 41.681 \times 10^6 \\ &= \underline{\underline{109.686 \times 10^6 \text{ mm}^4}} \end{aligned}$$

Moment of Inertia about y-y axis (I_{yy})

$$I_{yy} = I_{yy_1} + I_{yy_2} + I_{yy_3} \quad \text{--- (2)}$$

$$\begin{aligned} I_{yy_1} &= I_{G_1} + a_1 \bar{h}_1^2 \\ &= \frac{b_1 l_1^3}{12} + a_1 (\bar{x} - x_1)^2 \\ &= \frac{30(150)^3}{12} + 4500(0-0)^2 = 8.4375 \times 10^6 \text{ mm}^4 \end{aligned}$$

$$\begin{aligned} I_{yy_2} &= I_{G_2} + a_2 (\bar{x} - x_2)^2 = \frac{l_2 b_2^3}{12} + a_2 (\bar{x} - x_2)^2 \\ &= \frac{150(30)^3}{12} + 4500(0-0)^2 = 0.3375 \times 10^6 \text{ mm}^4 \end{aligned}$$

$$\begin{aligned} I_{yy_3} &= I_{G_3} + a_3 (\bar{x} - x_3)^2 \\ &= \frac{b_3 l_3^3}{12} + a_3 (\bar{x} - x_3)^2 \\ &= \frac{30(300)^3}{12} + 9000(0-0)^2 \\ &= 67.500 \times 10^6 \text{ mm}^4 \end{aligned}$$

Sub. in (2),

$$\begin{aligned} I_{yy} &= 8.4375 \times 10^6 + 0.3375 \times 10^6 + 67.500 \times 10^6 \\ &= 76.274 \times 10^6 \text{ mm}^4 \end{aligned}$$

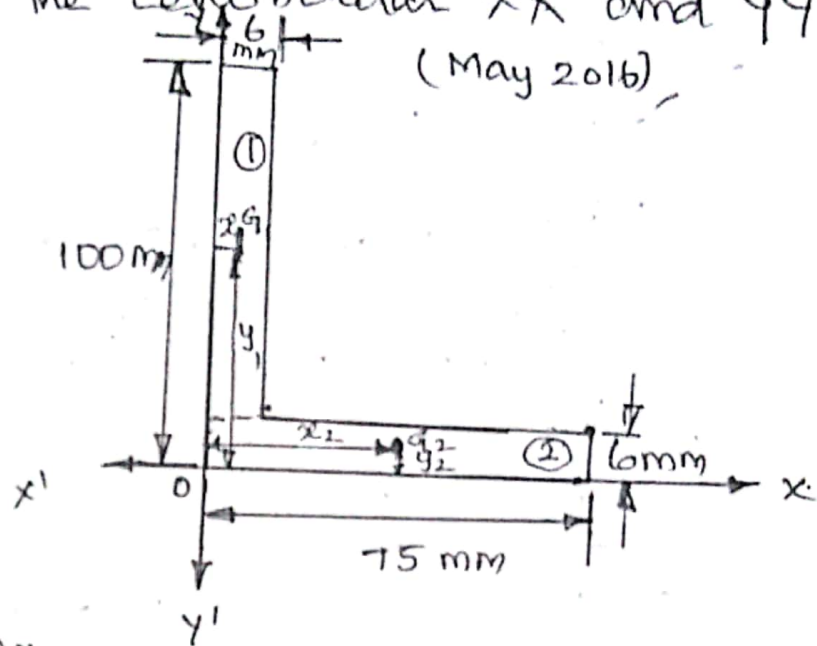
Polar Moment of Inertia (I_{zz})

$$\begin{aligned} I_{zz} &= I_{xx} + I_{yy} \\ &= 109.686 \times 10^6 + 76.274 \times 10^6 \\ &= 185.96 \times 10^6 \text{ mm}^4 \end{aligned}$$

5. Find the moment of inertia of 'L' angle ISA100x75 x 6 about the centroidal xx and yy.

Solution:

(May 2016)



Rectangle ①:

$$\begin{aligned}
 a_1 &= l_1 \times b_1 \\
 &= 94 \times 6 = 564 \text{ mm}^2 \\
 x_1 &= \frac{b_1}{2} = \frac{6}{2} = 3 \text{ mm} \\
 y_1 &= \frac{l_1}{2} + 6 \\
 &= \frac{94}{2} + 6 = 53 \text{ mm}
 \end{aligned}$$

Rectangle ②:

$$\begin{aligned}
 a_2 &= l_2 \times b_2 \\
 &= 75 \times 6 = 450 \text{ mm}^2 \\
 x_2 &= \frac{l_2}{2} = \frac{75}{2} = 37.5 \text{ mm} \\
 y_2 &= \frac{b_2}{2} = \frac{6}{2} = 3 \text{ mm}
 \end{aligned}$$

$$\bar{y} = \frac{a_1 y_1 + a_2 y_2}{a_1 + a_2}$$

$$\bar{x} = \frac{a_1 x_1 + a_2 x_2}{a_1 + a_2} = \frac{(564 \times 3) + (450 \times 37.5)}{564 + 450} = 18.31 \text{ mm}$$

$$\bar{y} = \frac{(564 \times 53) + (450 \times 3)}{564 + 450} = 30.31 \text{ mm}$$

Moment of Inertia about x-x axis,

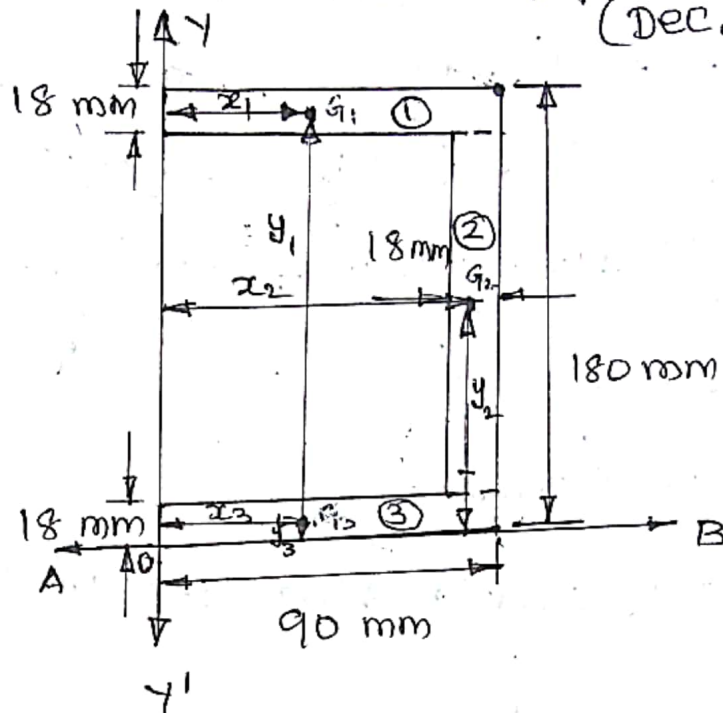
$$\begin{aligned} I_{xx} &= I_{xx_1} + I_{xx_2} \\ &= IG_1 + a_1 h_1^2 + IG_2 + a_2 h_2^2 \\ &= \frac{b_1 l_1^3}{12} + a_1 (y_1 - \bar{y})^2 + \frac{l_1 b_1^3}{12} + a_2 (\bar{y} - y_2)^2 \\ &= \frac{6(94)^3}{12} + 564(53 - 30.81)^2 + \frac{75(6)^3}{12} + 450(30.81 - 3)^2 \\ &= \underline{\underline{1.041 \times 10^6 \text{ mm}^4}} \end{aligned}$$

Moment of Inertia about y-y axis,

$$\begin{aligned} I_{yy} &= I_{yy_1} + I_{yy_2} \\ &= IG_1 + a_1 (\bar{x}_1)^2 + IG_2 + a_2 (\bar{x}_2)^2 \\ &= \frac{l_1 b_1^3}{12} + a_1 (\bar{x} - x_1)^2 + \frac{b_2 l_2^3}{12} + a_2 (x_2 - \bar{x})^2 \\ &= \frac{94(6)^3}{12} + 564(18.31 - 3)^2 + \frac{6(75)^3}{12} + 450(37.5 - 18.31)^2 \\ &= \underline{\underline{0.498 \times 10^6 \text{ mm}^4}} \end{aligned}$$

4. Determine the moments of inertia of the area shown with respect to the centroidal axes parallel and perpendicular to the side AB. (Dec. 2017)

Solution:



Rectangle ①:

$$a_1 = l_1 \times b_1$$

$$= 90 \times 18 = 1620 \text{ mm}^2$$

$$x_1 = \frac{l_1}{2} = \frac{90}{2} = 45 \text{ mm}$$

$$y_1 = \frac{b_1}{2} + 162 = \frac{18}{2} + 162 = 171 \text{ mm}$$

Rectangle ②:

$$a_2 = l_2 \times b_2$$

$$= 144 \times 18 = 2592 \text{ mm}^2$$

$$x_2 = \frac{18}{2} + 72 = 81 \text{ mm}$$

$$y_2 = \frac{l_2}{2} + 18$$

$$= \frac{144}{2} + 18 = 90 \text{ mm}$$

Rectangle ③:

$$a_3 = l_3 \times b_3 = 90 \times 18 = 1620 \text{ mm}^2$$

$$x_3 = \frac{l_3}{2} = \frac{90}{2} = 45 \text{ mm}$$

$$y_3 = \frac{b_3}{2} = \frac{18}{2} = 9 \text{ mm}$$

$$\bar{x} = \frac{a_1 x_1 + a_2 x_2 + a_3 x_3}{a_1 + a_2 + a_3}$$

$$= \frac{(1620 \times 45) + (2592 \times 81) + (1620 \times 45)}{1620 + 2592 + 1620}$$

$$= 61 \text{ mm}$$

$$\bar{y} = \frac{a_1 y_1 + a_2 y_2 + a_3 y_3}{a_1 + a_2 + a_3}$$

$$= \frac{(1620 \times 171) + (2592 \times 90) + (1620 \times 9)}{1620 + 2592 + 1620}$$

$$\bar{y} = 90 \text{ mm}$$

Moment of Inertia about AB (I_{AB})

$$\begin{aligned} I_{AB} &= I_{AB_1} + I_{AB_2} + I_{AB_3} \\ &= I_{G_1} + a_1 h_1^2 + I_{G_2} + a_2 h_2^2 + I_{G_3} + a_3 h_3^2 \\ &= \frac{l_1 b_1^3}{12} + a_1 (y_1 - \bar{y})^2 + \frac{l_2 b_2^3}{12} + a_2 (y_2 - \bar{y})^2 + \frac{l_3 b_3^3}{12} + a_3 (y_3 - \bar{y})^2 \\ &= \frac{90(18)^3}{12} + 1620(171 - 90)^2 + \frac{18(90)^3}{12} + 2592(90 - 90)^2 + \frac{90(18)^3}{12} + 1620(90 - 9)^2 \\ &= 25.81 \times 10^6 \text{ mm}^4 \end{aligned}$$

Moment of Inertia about y-y axis (perpendicular to side AB)

$$\begin{aligned} I_{yy} &= I_{yy_1} + I_{yy_2} + I_{yy_3} \\ &= I_{G_1} + a_1 (\bar{h}_1)^2 + I_{G_2} + a_2 (\bar{h}_2)^2 + I_{G_3} + a_3 (\bar{h}_3)^2 \\ &= \frac{b_1 l_1^3}{12} + a_1 (\bar{x} - x_1)^2 + \frac{l_2 b_2^3}{12} + a_2 (x_2 - \bar{x})^2 + \frac{b_3 l_3^3}{12} + a_3 (\bar{x} - x_3)^2 \\ &= \frac{18 \times (90)^3}{12} + 1620(61 - 45)^2 + \frac{90(18)^3}{12} + 2592(81 - 61)^2 + \frac{18(90)^3}{12} + 1620(61 - 45)^2 \\ &= 4.12 \times 10^6 \text{ mm}^4 \end{aligned}$$